

TrES-2b

R-1591

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130810 · created 2026-08-02T01:04:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456514.72076 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.084 +/- 0.04
Transit depth (Rp/Rs)^2	0.7 +/- 0.68 [%]
Orbital Inclination (inc)	84.27 +/- 2.89
Airmass coefficient 1 (a1)	0.929 +/- 0.016
Airmass coefficient 2 (a2)	0.07 +/- 0.17
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	#2 - [592, 163]
Optimal Method	PSF photometry
Transit Duration (day)	0.061 +/- 0.04

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.084 ± 0.04 vs 0.1254 (deviation 1.04σ)
Duration fit vs published	0.061 d vs 0.0746 d (deviation 0.34σ)
Mid-time O–C	0.6 min
Depth SNR	2.1
Residual scatter	1.69 %

Light curve

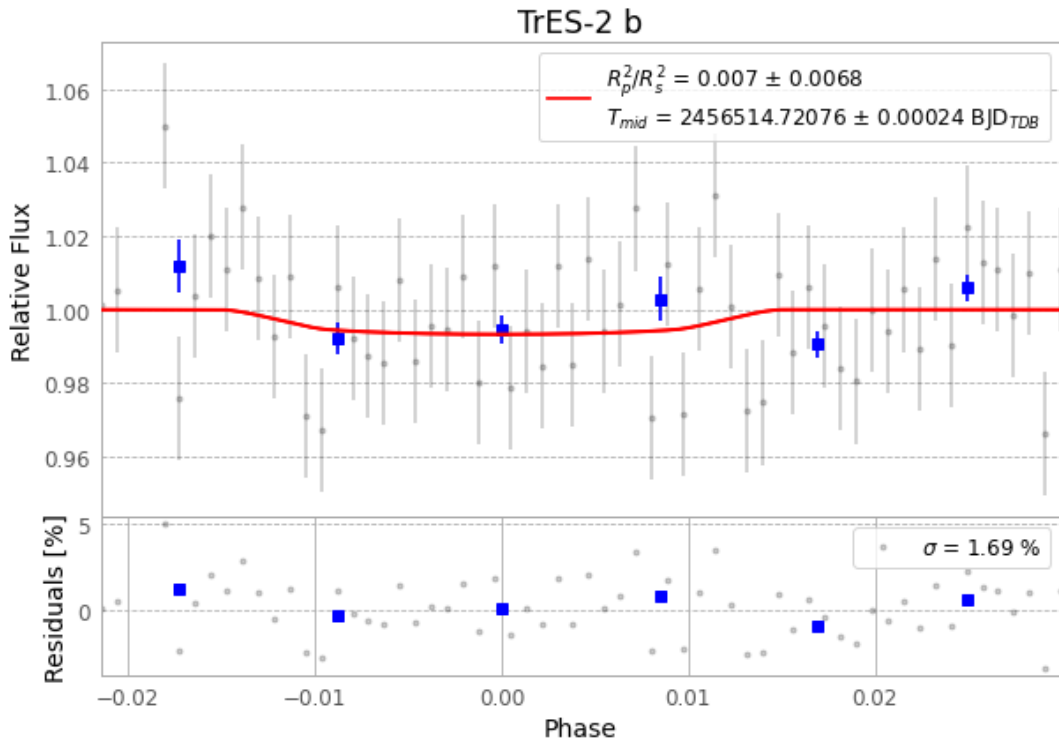


Figure 1 — FinalLightCurve_TrES-2 b_2013-08-09.png (SHA-256 e747585a2023e341...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 281], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 24e23ee6565c6a10...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), TRES-2130810035712.FITS → TRES-2130810070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130810.log (4ea5d9b60097...), FinalLightCurve_TrES-2 b_2013-08-09.png (e747585a2023...), FinalLightCurve_TrES-2 b_2013-08-09.csv (fb8c58c37629...)

Compute resources

Wall time 450.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:04Z.